Virtual University of Pakistan

BT402

Final term past papers solved

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by Zareen Fatima
mcqs

1. Microorganisms came into being on earth over a period of about **1.2 to 1.5** billion years
2. Which is not microorganism **liverfluck**
3. Shear Sensitivity is a **physiological factor**
4. GMOs, are commonly used in foods and **medicines**
5. Biotechnology is about **all**
6. The molecular characterization of all genes in a species. Is **genomics**
7. the **yeast** is used for fermenting malted cereals and fruit juices to produce **ethanol**.
8. The most common types of GMO's plants **canola** ,
9. Production of antibiotics, organic acids and enzymes by **fermentation** of microbes.
10. **Biopesticide** is a formulation made from naturally occurring substances that controls pests by **non toxic** mechanisms and in ecofriendly manner
11. In Conjugate Vaccines a bacterium possesses an outer coating of sugar molecules called **polysaccharides**,
12. A **vaccine** is a biological preparation that improves immunity to a particular disease.
13. **Bioremediation** is a natural process that involves the use of biological entities to neutralize the contaminated site
14. First-generationbiofuels are made from sugar, starch, **vegetable oil**, or animal fats using conventional technology
15. Seedling root dip This method is applied to the **rice** crop

Bt402 11:00am Final term 23 Feb, 2019 (by Rania Ch)

1. Define compost

Ans: Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment

• Compost is a key ingredient in organic farming.

• End product of the decomposition of organic matter

Composting

• Composting is nature's process of recycling decomposed organic materials into a rich soil known as compost

• Composting is a natural process that turns organic material into a dark rich substance

2. DNA vaccine with example 2 marks

- Still in the experimental stages, these vaccines show great promise, and several types are being tested in humans.
- • DNA vaccines take immunization to a new technological level.

- • These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe's genetic material.
- • Example: Influenza vaccine.

3. What is biochemical pesticides 2 marks

Ans:
• They are naturally occurring substance to control pest by non-toxic mechanisms.
• Biochemical pesticides include substances as insect sex pheromones, that interfere with mating that attract insect pest to traps

4. Define biosurfactants 2 marks

Ans:
• These are microbial-synthesized surface-active substances that allow for more efficient microbial biodegradation of hydrocarbons in bioremediation processes.

• There are two ways by which biosurfactants are involved in bioremediation
1) Increase the surface area of hydrophobic water-insoluble substrates. Growth of microbes on hydrocarbons can be limited by available surface area of the water-oil interface. Emulsifiers produced by microbes can break up oil into smaller droplets, effectively increasing the available surface area.
2) Increase the bioavailability of hydrophobic water-insoluble substrates. Biosurfactants can enhance the availability of bound substrates by desorbing them from surfaces (e.g. soil) or by increasing their apparent solubility. Some biosurfactants have low critical micelle concentrations (CMCs), a property which increases the apparent solubility of hydrocarbons by sequestering hydrophobic molecules into the centres of micelles.

5. DNA vaccines with example 3 marks

- Ans: Still in the experimental stages, these vaccines show great promise, and several types are being tested in humans.
- • DNA vaccines take immunization to a new technological level.
- • These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe's genetic material.
- • Example: Influenza vaccine.

6. What is biodiesel and how it is produced 3 marks

Ans: • **Biodiesel** is produced, mainly in the European Union, by combining vegetable oil or animal fat with an alcohol
• Biodiesel can be blended with traditional diesel fuel or burned in its pure form in compression ignition engines. Its energy content is somewhat less than that of diesel (88 to 95%).
• Biodiesel can be derived from a wide range of oils, including rapeseed, soybean, palm, coconut or jatropha oils and therefore the resulting fuels can display a greater variety of physical properties than ethanol.

7. How biofuel generate electricity 3 marks

Ans: Charging Electronics
• According to scientists from Saint Luis University, a fuel cell was developed with cooking oil and sugar to generate electricity
• consumers will be able to use these cells instead of generating electricity.
• Consumers may be able to use fuel cells in place of batteries to charge anything from computers to cell phones

8. Name steps of vaccine purification 5 marks

Ans: Steps of vaccine production

1. Generation of the Antigen VIRUS BACTERIA
2. Isolation of the Antigen
3. Purification
4. Addition of Other Components
5. Packaging

Generation of the Antigen VIRUS BACTERIA:

- The first step in order to produce a vaccine is generating the antigen that will trigger the immune response. For this purpose the pathogen's proteins or DNA need to be grown and harvested using the following mechanisms:
- Viruses are grown on primary cells such as cells from chicken embryos or using fertilised eggs (e.g. influenza vaccine) or cell lines that reproduce repeatedly (e.g. hepatitis A)
- Bacteria are grown in bioreactors which are devices that use a particular growth medium that optimises the production of the antigens

- Recombinant proteins derived from the pathogen can be generated either in yeast, bacteria or cell cultures.

Isolation of the Antigen:

- The aim of this second step is to release as much virus or bacteria as possible.
- To achieve this, the antigen will be separated from the cells and isolated from the proteins and other parts of the growth medium that are still present.

Purification:

- In a third step the antigen will need to be purified in order to produce a high purity/quality product.
- This will be accomplished using different techniques for protein purification. For this purpose several separation steps will be carried out using the differences in for instance protein size, physico-chemical properties, binding affinity or biological activity.

Addition of Other Components:

- The fourth step may include the addition of an adjuvant, which is a material that enhances the recipient's immune response to a supplied antigen. The vaccine is then formulated by adding stabilizers to prolong the storage life or preservatives to allow multi-dose vials to be used safely as needed.
- Due to potential incompatibilities and interactions between antigens and other ingredients, combination vaccines will be more challenging to develop. Finally, all components that constitute the final vaccine are combined and mixed uniformly in a single vial or syringe.

9. Advantages of biopesticides 5 marks

• Inherently less harmful and less environmental load,

• Designed to affect only one specific pest or, in some cases, a few target organisms,

• Often effective in very small quantities and often decompose quickly, thereby resulting in lower exposures and largely avoiding the pollution problems .

• When used as a component of Integrated Pest Management (IPM) programs, biopesticides can contribute greatly.

10. Role of microbes in petroleum industry 10 marks

Ans: **Role of microbes in petroleum industry**

• Petroleum-based products are the major source of energy for industry and daily life. Leaks and accidental spills occur regularly during the exploration, production, refining, transport, and storage of petroleum and petroleum products.
• Release of hydrocarbons into the environment whether accidentally or due to human activities is a main cause of water and soil pollution
• Soil contamination with hydrocarbons causes extensive damage of local system since accumulation of pollutants in animals and plant tissue may cause death or mutations
• Microbial processes play an important role in industry, the protection of the natural environment
• Several practical applications of microbial processes are listed as follows.

Bioremediation
• **Bioremediation** is a natural process that involves the use of biological entities to neutralize the contaminated site
• **Microbial** biodegradation is the use of **bioremediation** and biotransformation methods to harness the naturally occurring ability of **microbial** xenobiotic metabolism to degrade, transform or accumulate environmental pollutants, including hydrocarbons (e.g. oil), polychlorinated biphenyls (PCBs), polyaromatic hydrocarbons
Biosurfactants
• These are microbial-synthesized surface-active substances that allow for more efficient microbial biodegradation of hydrocarbons in bioremediation processes.
• There are two ways by which biosurfactants are involved in bioremediation

1) Increase the surface area of hydrophobic water-insoluble substrates. Growth of microbes on hydrocarbons can be limited by available surface area of the water-oil interface. Emulsifiers produced by microbes can break up oil into smaller droplets, effectively increasing the available surface area.
• 2) Increase the bioavailability of hydrophobic water-insoluble substrates. Biosurfactants can enhance the availability of bound substrates by desorbing them from surfaces (e.g. soil) or by increasing their apparent solubility. Some biosurfactants have low critical micelle concentrations (CMCs), a property which increases the apparent solubility of hydrocarbons by sequestering hydrophobic molecules into the centres of micelles.
• Microbial Enhanced Oil Recovery (MEOR)
• One of the major concerns facing the oil industry today is the recovery of the large percentage of oil remaining unrecovered in mature and in nearly depleted oil fields.
• Microbes can influence and improve the oil recovery process by:
• generating gases that increase reservoir pressure and decrease oil viscosity
• generating acids that dissolve rock, thus improving absolute permeability
• reducing permeability in channels

• producing bio-surfactants that decrease interfacial tension
• reducing oil viscosity by degrading long-chain saturated hydrocarbons.

Define vaccine (2)

• **Vaccines** are a product of **Industrial Microbiology**. Many antiviral vaccines are massproduced in chicken eggs or cell cultures.
• The production of vaccines against bacterial diseases usually requires the growth of large amounts of the bacteria. **Recombinant DNA technology** is increasingly important in the development and production of subunit vaccines.

What is soil treatment (2)

• All the bio fertilizers along with the compost fertilizers are mixed together.

• They are kept for one night. Then the next day the mixture is spread on the soil where seeds have to be sown

Role of antimicrobial pesticides (5)

Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centers, public access facilities and homes

Challenges in adjuvant (5)

Challegnes with adjuvant
• Vaccines containing novel adjuvant formulations are increasingly reaching advanced development and licensing stages, providing new tools to fill previously unmet clinical needs.

• However, many adjuvants fail during product development owing to factors such as manufacturability, stability, lack of effectiveness, unacceptable levels of tolerability or safety concerns.

• Lower reactogenicity adjuvants came with reduced vaccine efficacy.

• Safety and tolerability are critical regulatory issues facing the adjuvants.
• In most cases increased adjuvant potency is associated with increased reactogenicity and toxicity.

3: Live attenuated vaccines

An **attenuated vaccine** is a **vaccine** created by reducing the virulence of a pathogen, but still keeping it viable (or "**live**"). **Attenuation** takes an infectious agent and alters it so that it becomes harmless or less virulent. These **vaccines** contrast to those produced by "killing" the virus (**inactivated vaccine**).

4: Role of biotechnology in agriculture

Agricultural biotechnology offers efficient and cost-effective means to produce a diverse array of novel, value-added products and tools.
• It has the potential to increase food production, reduce the dependency of agriculture on chemicals, and lower the cost of raw materials, all in an environmentally friendly manner.
• The initial phase of a revolution in agriculture has already occurred. Large areas of genetically modified (GM) crops of soybeans, maize, cotton, and canola have been successfully grown in the Western Hemisphere.

The main objectives of creating transgenic plants are attempts to engineer metabolic pathways for the production of tailor-made plant polymers or low molecular weight compounds, increased resistance towards pathogens and pesticides, improved food quality, and the production of polypeptides for pharmaceutical or technical use. Plantmade vaccines or antibodies are especially attractive as plants are free of human diseases, reducing screening

5: which type of relationship between plants and microbes

Plant-microbe interactions
• Plant-microbe interactions describe a broad range of scientific studies concerning how microbes interact with plants at the molecular biology and molecular genetics level.

• Plants and microbes can have a variety of interactions including

• Symbiotic

• associative

• pathogenic,

• all of which impact plant productivity, stress tolerance and disease resistance.

• Root Associations

• The enzyme system for nitrogen fixation is found only in the prokaryotes, and in the case of symbiotic nitrogen fixation in plants,

• there is considerable specificity between the legume symbiont and bacteria for the stable association.

• The plant provides the carbon and energy source for the bacteria to grow, while the bacteria fix nitrogen with the production of amino acids for plant growth.
Pathogenic interactions
• Bacterial Pathogens

• The bacteria-producing diseases in plants generally display a number of hydrolyticenzymes for penetration of the plant surface, and growth in the plant is generally intercellular between the parenchyma cells. ›

• As the disease progresses in the plant, bacterial distribution may become systemic as a result of migration into the vascular tissue.

• In some cases, the bacteria release pectinase that hydrolyzes the plant cell walls and the plant collapses.

6: Biofuels

A biofuel is a fuel that is produced through contemporary biological processes, such as agriculture and anaerobic digestion, rather than a fuel produced by geological processes such as those involved in the formation of fossil fuels, such as coal and petroleum, from prehistoric biological matter.

7: Role of microbes in oil recovery process (5)

• One of the major concerns facing the oil industry today is the recovery of the large percentage of oil remaining unrecovered in mature and in nearly depleted oil fields.

• Microbes can influence and improve the oil recovery process by:

• generating gases that increase reservoir pressure and decrease oil viscosity

• generating acids that dissolve rock, thus improving absolute permeability

• reducing permeability in channels

• producing bio-surfactants that decrease interfacial tension

• reducing oil viscosity by degrading long-chain saturated hydrocarbons

8: Main object in transgenic plants (5)

The main objectives of creating transgenic plants are attempts to engineer metabolic pathways for the production of tailor-made plant polymers or low molecular weight compounds, increased resistance towards pathogens and pesticides, improved food quality, and the production of polypeptides for pharmaceutical or technical use.

1. **Benefits of compost**

Composting Benefits
• **Soil conditioner**: With compost, you are creating rich humus for lawn and garden. This adds nutrients to your plants and helps retain moisture in the soil.

• **Recycles kitchen and yard waste**: Composting can divert as much as 30% of household waste away from the garbage can.

• **Introduces beneficial organisms to the soil**: Microscopic organisms in compost help aerate the soil, break down organic material for plant use and ward off plant disease.

• **Good for the environment**: Composting offers a natural alternative to chemical fertilizers.

• **Reduces landfill waste**: Most landfills in North America are quickly filling up; many have already closed down. One-third of landfill waste is made up of compostable materials

Nitrogen bio fertilizers?

• The sort of bio fertilizers can help any farmers to determine the nitrogen point during the solid ground. Nitrogen may be a crucial factor which happens to be put to use in any growing for put.

• Including, Azotobacteria must be used to your non-legume bounty; Rhizobium it takes to your legume bounty. Purple earth-friendly algae are needed to progress hemp despite the fact that Acetobacter must be used to progress sugarcane

Steroids ?

• Steroids are a very important group of chemicals, which are used as anti-inflammatory drugs, and as hormones such as estrogens and progesterone, which are used in oral contraceptives.

• Producing steroids from animal sources or chemically systhesising them is difficult, but microorganisms can synthesise steroids from sterols or from related compounds

Worm composting?

Worm composting is using worms to recycle food scraps and other organic material into a valuable soil amendment called vermicompost, or worm compost.

What is Biofuel second generation?

Second-generation

• **Second-generation** biofuels are produced from non-food crops, such as cellulosic biofuels and waste biomass (stalks of wheat and corn, and wood).
• Common second-generation biofuels include vegetable oils, biodiesel, bioalcohols, biogas, solid biofuels, and syngas.
• Research continues on second-generation biofuels including biohydrogen, biomethanol, DMF, Bio-DME, Fischer-Tropsch diesel, biohydrogen diesel, mixed alcohols and wood diesel.

Subunits Vaccines

• Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the
immune system.
• In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies
or T cells recognize and bind to. immunization.
• Because subunit vaccines contain only the essential antigens and not all the other molecules that make up the microbe.
• Example: Plague immunization.
• Toxoid Vaccines
• For bacteria that secrete toxins, or harmful chemicals, a toxoid vaccine might be the answer.

steps in transgeneic plants?

Genetic modification
• involves the
• mutation
• insertion
• or deletion of genes.
• Inserted genes usually come from a different species in a form of horizontal gene transfer
• In nature this can occur when exogenous DNA penetrates the cell membrane for any reason

This can be accomplished artificially by
• attaching the genes to a virus.
• physically inserting the extra DNA into the nucleus of the intended host with a very small syringe.
• using electroporation (that is, introducing DNA from one organism into the cell of another by use of an electric pulse).
• firing small particles from a gene gun